

FIFE NESS, United Kingdom

WMO: 031740

Lat: 56.279N

Lon: 2.587W

Elev: 12

StdP: 101.18

Time Zone: 0.00 (EUW)

Period: 94-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.6	0.4	-4.6	2.6	1.1	-3.5	2.8	1.7	22.6	4.5	20.7	4.1	5.3	290	0.768

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.7	20.3	16.1	19.0	15.4	17.9	14.8	17.0	19.1	16.2	17.9	15.6	17.1	6.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.4	18.0	15.4	11.0	17.2	14.8	10.5	16.5	47.7	19.3	45.4	18.0	43.7	17.3	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.3	13.7	DB	-2.7	23.2	1.7	2.1	-4.0	24.7	-5.0	25.9	-6.0	27.1	-7.3	28.6	(4)
(5)			WB	-3.4	18.9	1.4	1.0	-4.4	19.6	-5.2	20.2	-6.0	20.8	-7.0	21.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.5	5.3	5.2	6.1	7.7	10.0	12.7	14.6	15.0	13.4	10.7	7.8	5.6		
	DBStd	4.16	2.03	2.21	2.32	2.14	2.10	1.89	1.77	1.53	1.75	2.27	2.45	2.71		
	HDD10.0	734	147	134	123	75	25	1	0	0	1	19	71	138		
	HDD18.3	3211	405	367	380	319	257	169	116	104	148	236	314	396		
	CDD10.0	567	0	1	1	6	26	82	144	154	103	42	6	1		
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH23.3	4	0	0	0	0	0	0	3	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.4	7.9	7.9	7.0	6.3	5.5	5.4	4.5	4.8	5.7	7.3	7.5	7.6		
(15) Precipitation	PrecAvg	670	57	50	44	45	54	64	50	57	53	79	63	58		
	PrecMax	940	111	99	114	98	97	108	116	171	121	148	108	98		
	PrecMin	486	19	7	17	4	17	26	21	12	15	13	35	22		
	PrecStd	107	24	28	25	28	23	24	25	40	32	41	20	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	12.4	13.6	16.0	18.3	20.4	22.9	22.2	20.4	16.9	13.7	12.6		
		MCWB	9.8	11.2	10.7	11.7	13.8	15.8	18.4	17.3	16.4	13.8	12.2	11.7		
	2%	DB	10.3	10.9	11.9	13.8	16.2	18.7	20.4	20.1	18.4	15.5	12.7	11.0		
		MCWB	9.0	9.4	9.5	10.6	12.4	14.9	16.2	16.2	15.1	13.1	11.3	9.7		
	5%	DB	9.4	9.8	10.7	12.3	14.4	17.2	18.8	18.7	17.2	14.6	11.9	10.1		
		MCWB	8.0	8.4	8.9	9.6	11.5	13.8	15.3	15.3	14.4	12.6	10.6	9.0		
	10%	DB	8.5	8.6	9.6	11.0	13.0	16.0	17.7	17.6	16.1	13.7	11.2	9.2		
		MCWB	7.2	7.1	8.0	8.9	10.9	13.2	14.8	15.0	13.8	12.1	10.1	8.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	11.4	11.4	12.5	14.7	16.6	18.7	18.2	17.5	14.6	12.6	11.6		
		MCDB	11.1	12.3	12.9	15.0	17.0	19.0	22.7	21.2	19.7	16.1	13.2	12.3		
	2%	WB	9.2	9.7	10.1	11.1	13.1	15.4	17.0	17.0	15.8	13.8	11.6	9.9		
		MCDB	10.2	10.7	11.3	13.2	15.2	18.0	19.4	18.8	17.4	14.9	12.3	10.7		
	5%	WB	8.2	8.5	9.1	10.1	12.1	14.3	16.1	16.3	14.9	13.1	11.0	9.1		
		MCDB	9.1	9.7	10.4	11.9	13.9	16.5	18.0	17.7	16.5	14.1	11.6	10.0		
	10%	WB	7.4	7.4	8.1	9.1	11.2	13.5	15.3	15.7	14.2	12.5	10.4	8.3		
		MCDB	8.3	8.4	9.4	10.7	12.6	15.4	16.9	17.0	15.6	13.5	11.1	9.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.9	4.3	4.4	4.5	5.0	5.0	4.7	4.4	4.0	3.9	3.5		
		MCDBR	4.7	5.0	6.1	7.0	7.1	7.5	7.7	6.9	6.1	5.0	4.4	4.0		
	5% WB	MCWBR	4.5	4.5	4.7	4.5	4.6	4.6	4.6	4.1	4.1	3.8	4.1	3.9		
		MCWBR	4.5	5.0	5.9	6.5	6.2	6.7	6.6	5.4	5.2	4.2	3.9	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.334	0.365	0.386	0.382	0.384	0.385	0.373	0.358	0.342	0.315	0.298		
	taud		2.285	2.379	2.327	2.296	2.359	2.381	2.384	2.427	2.476	2.454	2.347	2.246		
	Ebn at Noon		574	729	792	830	857	859	849	837	799	712	576	491		
	Edh at Noon		53	71	95	113	112	111	109	98	82	66	49	43		
(44) All-Sky Solar Radiation	RadAvg		0.50	1.15	2.17	3.56	4.74	4.66	4.49	3.58	2.52	1.36	0.65	0.37		
	RadStd		0.07	0.11	0.27	0.38	0.30	0.40	0.40	0.25	0.17	0.16	0.07	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page